

# A Packing of 13 Four-Spaces in $\mathbb{R}^9$

Limin Ge\*

October 9, 2026

## Abstract

We give 13 4-dimensional subspaces of  $\mathbb{R}^9$  whose smallest squared chordal distance is at least 2.407101. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.392880, obtained by deleting 1 of the 14 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

## 1 Result

**Theorem.** *There are 13 subspaces of dimension 4 in  $\mathbb{R}^9$  whose pairwise squared chordal distances  $D(a, b) = 4 - \text{tr}(P_a P_b)$  are all at least*

$$D_{\min} = 2.407101 \dots$$

The exact value of  $D_{\min}$ , a rational number, is given in Appendix A.

## 2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some  $N$  in the row  $m = 9$ ,  $n = 4$ . The cell  $N = 13$  is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.392880, obtained by deleting 1 of the 14 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is  $4(9 - 4)/9 \cdot 13/12 = 2.407407$ , and the value above lies below it.

## 3 Verification

Each subspace is given by an integer  $m \times n$  basis  $Y$  (coordinates times  $10^{12}$ , rounded). With  $G = Y^\top Y$  the projection is  $P = YG^{-1}Y^\top$ , so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

```
python grassmann/verify.py grassmann/packings/m9n4N13.txt
```

in <https://github.com/Ge-limin/math>.

---

\*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

## 4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 62 seconds on one laptop, with about ten such searches running at once.

## A Exact value

numerator:

```
120355063330612121592725085433619877117600247149688832890866523206031334
304242060077755180586611395186331388569918672561289215574836059995644035
8000222379217808890991672906739969989507956637399
```

denominator:

```
500000000000614669073304355776480310801550242028511440434606497579964032
305880636052574350024891023560351910245178941014094301518652866260680037
015180786441172319898429480746856487969717903523
```

## B Coordinates

13 blocks of 9 rows with 4 integers each; divide by  $10^{12}$ . File: `grassmann/packings/m9n4N13.txt`.

```
-98096102223 93851004525 -615724639330 604017186729651804536533 194554832996 85705036725 450847157602250966076771 -102728623152 471691538376 -71423711504
-663597627523 443556847453 297990397086 36987391545237780158059 -489785346352 -284685932685 2230961212365591206534 138106237849 -4832222012 339907579107
387608555851 231952434782 -76655498146 -28236634593317953038334 -371724077853 -9727753373 -8057406237364680894083 -230177372612 710525816440 -182905114837
-217962791620 -190256546626 225611134723 3111637177455141599227 -57628412971 -352804550990 5510334887837040345356 627181759907 -76740014061 -480748363674
-228696822942 -380639677039 -13033962623 221727100287611716470 -95975956369 599098281153 27376949100410764090193 40182489737 299371285606 542600233224
-102847734714 -310638993421 -214681616245 -3872003124232677148 178133835933 -153159646771 -557277593935213958927 -39488332181 603821252208 -106394984137
490582604174 112232756452 531020729915 45939451528732031983819 -272296302423 646189990578 -384513781391785566070 57520256344 -14567133298 -299982114002
-59175482194 -644257080941 296126310941 -772417575067492357922 42387272417 48036252506 268562627686 -361216795832 203457251254 -129845825257 -115476308460
-212636426334 188157648153 243673047531 -216913775336845443070 -170115188395 -327230758172 7295473115548487413414 377500670929 257465417258 84015079026
225478483824 -48602069184 -13614683163 -5448542961729091482307 -524466618028 88018375727 46268125479942733209574 384762573953 412826420138 262165721754
-211015601421 -655371213526 -79791998931 416421476733557469560 742940458573 199987463610 165346224720 323233280266 -248793781809 -215315941779 178526698200
675418642786 -89189787596 340659731678 243460591710188470507578 126482565679 561790444586 8786555694037123432294 -454222748736 489633036301 -502156228087
-414727715109 478518471447 517048498374 33749166739283337208963 170768608566 -145154194443 -11931148609560770484 -231669146313 -618699110304 -165341406505
142776629531 -180533679887 343434646672 -2026160959435585768357 -17856922435 -97992184781 -691483365335176800668 228782721709 -295049027237 133662041028
-331670915258 -390604670856 17842949444 -2369047988255980272421 -133303911032 281119393303 1821612613379578879167 -420576425741 384196601643 -83184431885
-99084132657 306629373483 -576999581841 -59841837914148027876285 194906555824 222029298817 1777040604081765294800 -465477532647 -236027834376 714870502246
50771219887 -142178116945 -234617667837 40643953047341127938150 -108652699016 691524835662 -1597203667420901619815 163645468707 59026625405 474885089723
365590350675 178300469815 -322211079831 3010015960730516531278 103106189604 231206590226 -17936655090501394513564 -408813448498 -411244311530 -342578835627
-2712039531 -79436322172 -207617845464 70018347365274298669053 -482309897075 -119156316169 -474759480530999249061 553614948111 -365594179569 23402860060
-225537189371 534400604222 138657202826 180826136362476412047 609680289077 -403035332824 -15708226875385095329897 -3695253064 -143340239664 166920634003
-38477965390 157523693322 -712671033699 8939927342 -297339852559 -374313098094 -110941833759 691331367272625774104 4750667065 -5455309920 -256029089009
441343212391 -396203173418 -293308051624 16563101338670712057889 304711328163 387280903418 -851275232448374592706 93209409732 182769792474 310742930656
-557674972809 -385216711728 268507768805 3529270779667341549939 -224473425217 -264881930387 -449338103300671356523 212740299119 -259294361944 63936186547
-19927381454 456824982035 -254360999555 109473973962110958434 227803950600 -99735509735 109606551923 -150609067763 -606679059849 228208444550 -271179803104
-505309198497 -308598071757 -379595094341 -471984593666481081 -312033243621 -173191928688 -314804911407157110742 -18673611022 -234785839216 441117622145
-430348606814 156156098396 -187101442832 142992385451495190598 700847343916 165361032749 -395382300942809579082762 -560617080270 -324098415622 560203940848
-30421222951 -222748109261 -166865422926 2641459908549016034101 224203035671 206451557446 3276230281324453191850 76986577708 -213587316212 179837893228
539804607853 -364627679765 53776292039 -2561791743067828632635 -107333467865 -106538044325 1068151555550325188626 -31820766326 629646432517 477204785824
128366284777 -13244133572 305990216695 -30316643644886618251325 233331100520 175998221105 -3818057927455713737235 -215408112761 422733440843 -38188819460
546739163591 203307637393 -475321705119 -1728807026331648085030 512099819093 -577208262519 2164728037435487989886 -458272545119 -257566759735 -236378586577
-276568487180 552797687310 -42213865591 -9214770118429141453661 85932971036 -546025770014 21903425147388013458765 586601858985 -43434513696 335424998864
-190011955808 211865443815 -85284938768 -607361646622070431037 -25997396764 40136289124 314348905407 -291083840081 246252946350 -7637638924 413172126326
67894445220 173686210895 -680617395982 25049009351678068689383 -157831598666 -474117990428 -53516072533091633407 406671940330 -253757899231 200468721799
-304365728222 13397986850 -58613218550 23644537052 82735229901 -404303124152 -9945668511 -13097522722322978552697 279474834671 425741250349 -339639063887
395363379743 512471243499 423911533669 468509233204747283462876 212708914989 172032197917 -1300920119324182036 265681644491 -434058243545 -215672489870
164482046344 426722183575 148712156985 -3903715970662294784160 -249242336160 57727874720 771820047898326497648072 425340325347 343163034400 -547099601914
254314344148 200073431613 428587155520 111208296003268547962529 -88516493437 236553493373 4440329059756354737133 140536838978 421041392502 305686475009
-511712022138 -195725967930 449705933692 -12920709230655654904 735103862434 372506052846 127826406029339144621013 266453051657 -409954830618 -236844441180
-140258852616 -679321593323 136908295849 -1492148353870675594 313609909227 -211877301357 51164150925 -618471686753 97573227625 317647889189 256640515905
```